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Deep Learning-Based Secure Real Time Face Recognition System

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Abstract: In that era of digital transformation, all day-to-day activities both in households and organizations were getting automated. In any organization, maintaining and monitoring attendance was an important activity. However, many challenges, such as using counterfeit identities or fabricating attendance records, existed in the attendance monitoring system. Facial features are important biological features because they serve as the initial point of recognition in our social relationships. In this work, a Deep Learning (DL) based application was designed and implemented to capture the attendance and record the presence of an employee or student in real time. This work proposed the concept of developing a face recognition application using the Python framework, which employed Deep Learning algorithms such as Convolutional Neural Network (CNN) to extract facial features and recognize faces of individuals and Viola Jones' Haar cascade technique was used to detect live faces. Security concerns such as spoofing attacks were mitigated by employing the Chaotic Logistic Key Mapping Algorithm along with the PyQt framework for facial encryption. The work proposed used PostgreSQL to store information about the users. It is interesting to note that the combined use of two algorithms, namely CNN and Viola Jones' Haar cascade technique, improved the prediction accuracy. The findings demonstrated the application's efficacy in facial recognition, live face detection, and spoofing attack protection, providing a comprehensive solution for robust face recognition. The proposed face recognition system employed several machine learning algorithms such as Eigen Faces, LBPH algorithm, SVM Classifier, CNN, and performance metrics of these algorithms were presented. Performance of ML algorithms was compared based on accuracy, precision, recall, and F1 score. It was observed that CNN provided better results when compared to other algorithms. The CNN model was tested with different data split ratios of 60:40, 70:30, 80:20, and 90:10, and it was discovered that the 90:10 data split ratio resulted in better prediction accuracy.

Keywords: Facial Recognition, Viola Jones, Convolutional Neural Network, Chaotic Logistic Mapping, PyQt

1 Introduction

Human faces played a crucial part in our daily lives, especially in identifying individuals. Face recognition was a biometric identification method that extracted and stored a person's distinct facial characteristics in order to identify them. Advancements in artificial intelligence and computer vision, particularly facial recognition, resulted in novel solutions for detecting someone's presence. As the face was a unique biometric trait that varied from person to person, this technology was secure and difficult to impersonate. It could precisely identify people without requiring direct contact. Facial recognition technology offered advantages for use in intelligent attendance systems, where the likelihood of deceiving user information decreased and reliability increased. While facial recognition technology had many advantages, its accuracy in recognizing faces, especially for attendance systems, required further training and testing. It surpassed other biometric recognition techniques such as fingerprint, palmprint, and iris recognition due to its non-contact method. It recognized people from a distance without the need for direct contact. This technology was used to identify individuals in various applications, including social media, airports, train stations, and criminal investigations. A good example, a social media application, Facebook employed facial recognition technologies to automatically tag users. Face identification in stationary and moving photos was a common research issue in image processing and pattern recognition. It required a large dataset and complex features to identify a person under all conditions such as lighting change, age, pose, and so on. Most facial recognition techniques worked well only when there were fewer people in a single frame, the lighting was controlled, the faces were properly positioned, and the image was clear. Recognition required large datasets and complex features to distinguish between distinct persons while navigating various barriers such as illumination, position, and age. According to recent research, facial recognition systems had improved. The rise of deep learning paved the way for the solution of many problems that were previously considered difficult for regular machine learning algorithms to solve. CNNs, in particular, altered the approach of addressing computer vision and image processing difficulties by delivering efficient results. CNN-based facial recognition systems grew in popularity due to their effectiveness in ensuring authenticity by automatically detecting faces. Face recognition was a difficult and compelling problem that captivated researchers across disciplines, including psychology, pattern recognition, neural networks, computer vision, and computer graphics. The following techniques were popularly used for facial recognition.

Feature-based method: This approach extracted local features such as eyes, nose, and mouth, then fed these features into a structured classifier based on their location and local statistics such as geometric or appearance. Feature restoration was a major challenge for feature extraction methods because it entailed extracting features that were not apparent owing to large variations, such as head attitude when comparing a frontal image to a profile image.

Holistic Method: The holistic approach considered the entire face region as input data for a face detecting system. Some examples of holistic approaches included Eigenfaces [3], Principal Component Analysis, Linear Discriminant Analysis [4], and Independent Component Analysis. The process of facial recognition typically involved several key stages to identify individuals within a database. Through PCA, eigenfaces were derived from the image data, facilitating the representation of each image as a vector of weights. The closest match was returned, allowing the user to identify the individual accurately and efficiently across diverse datasets.

Hybrid Method: Hybrid face recognition systems employed both holistic and feature extraction techniques, often leveraging 3D imaging for comprehensive analysis. By capturing facial structures in three dimensions, these systems captured intricate details such as eye socket curves and chin shapes, even from a profile view [29]. The process typically involved the following steps: Detection, where faces were captured via scanning or real-time photography; Position determination, to assess head location, size, and orientation; Measurement assignment, where precise metrics were assigned to facial curves, focusing on features such as eye contours and nose angles; Representation conversion, translating the measurements into a numerical code; and Matching, where the obtained data was compared with existing facial profiles in the database.

2Related Work

Algorithms used in facial recognition were broadly categorized into two groups: those enabling autonomous selection of optimal facial features and those where human intervention was necessary to select facial features [6]. Shrestha and Panday proposed that effective facial recognition could benefit from proficient feature extraction and categorization [5]. At that time, facial recognition found applications in various domains, such as monitoring student attendance in educational institutions [7, 8]. Humans effortlessly and successfully perceived faces, a task that proved challenging for computers. The human visual system employed intricate neural pathways to process both static and dynamic facial features, allowing for face recognition alongside contextual understanding.

CNN altered the approach to computer vision and image processing related issues and challenges by delivering efficient results. CNN-based facial recognition systems grew in popularity due to their effectiveness in ensuring authenticity by automatically detecting faces. Biometric applications used a range of static and dynamic biometric characteristics, such as voice, palm and finger vein patterns, gait analysis, DNA, hand geometry, iris patterns, and fingerprints.

Jayapriya et al. [28] presented an overview of template protection techniques: feature transformation or cancellable biometrics and bio-cryptosystems. Hybrid template protection schemes were created by combining these techniques to secure multimodal biometric templates. The work done in [28] provided a comprehensive understanding of template protection techniques and multimodal biometric systems for user authentication. Bruce and Young proposed a hypothesis suggesting that multiple independent subprocesses collaborated during face perception. These subprocesses extracted various attributes such as age, gender, and basic facial expressions from the face's physical features, forming a personalized face model structure. This structure aided the brain in recognizing faces even in diverse conditions. Some studies delved into identifying the most crucial facial features, such as eyes, mouth, and nose, for face recognition. The history of face recognition traced back to the 1950s and 1960s, with automated face recognition research gaining traction in the 1970s. Early methodologies relied on features based on distances between key facial regions. The 1990s marked a significant surge in face recognition research due to advancements in hardware and the heightened demand for security applications. The face recognition system, pioneered by Woodrow Wilson Bledsoe in 1966, used a facial database to match faces captured in digital images or videos. In educational institutions and universities, teachers and students saved time by using facial recognition technology, which also avoided fraudulent attendance [11]. According to Mansoor et al. [10], facial recognition involved filming student videos, converting them into frames, linking them to a database to verify their presence or absence, and annotating each student's presence so that records could be kept [9].

Obaida et al. conducted a comparative study on the Viola-Jones and YOLOv3 face detection algorithms [12]. It systematically covered the procedures involved in face identification using Viola-Jones, as well as a full algorithmic description of YOLOv3 and its design. The empirical data revealed that Viola-Jones attained a 45% accuracy rate, emphasizing its speed, while YOLOv3 demonstrated a substantially better accuracy rate of 90%. The comparative analysis presented the trade-offs between speed and accuracy, inspiring ideas for improving each algorithm or developing a hybrid technique that combined their merits. This research was consistent with the larger goal of building a face identification system that balanced speed and precision for practical applications. Kamgar-Parsi et al. proposed the primary goal of accurately recognizing and following individuals on a watchlist [13]. To improve the recognition system's precision, the study included novel approaches such as image morphing and automated exemplar generation and labelling. A critical step was customizing morph sets by analysing training sets, which contributed to model optimization. The evaluation criterion was based on the acceptance of test images projected within the decision region, with network output serving as an indicator. The system accepted test photographs when the activity of the output neuron hit a threshold of 0.5, demonstrating its dependability and efficiency in identifying individuals on the watchlist.

S. Garg et al. proposed a robust liveness detection technique for facial recognition that used LBPSVM, LBPDBN, and DeBNNet in [14]. This proposed approach in [14] used Deep Belief Networks (DBN) for deep feature extraction and classification, emphasizing a unified strategy for improving accuracy. The work not only addressed the difficulty of enhancing results across 2D and 3D face spoof datasets, but it also analysed cross-dataset scenarios, exhibiting a dedication to real-world application and generalizability. It was an advanced solution for maintaining the reliability and accuracy of face recognition systems in the presence of spoof attempts as the proposed work used integrated machine learning approaches and prevented several attacks. Andre Budiman et al. developed a university student attendance tracking system by replacing the current RFID technology with a more advanced mechanism [16]. The study compared two facial recognition algorithms, CNN and Local Binary Pattern Histogram (LBPH), to discover which provided higher accuracy in attendance monitoring. Specifically, LBPH obtained 77.55%, Eigenfaces 85%, PCA 80%, Haar Cascade 92.9%, and CNN achieved an amazing 95% accuracy. The findings indicated that CNN was the most effective algorithm for facial recognition in a variety of settings. The university enhanced the attendance tracking accuracy and efficiency using the methodology proposed in [16].

Bhoyar et al. presented an enhanced face detection and live face recognition technique that combined the Viola-Jones algorithm from OpenCV with the LBPH algorithm and used Support Vector Machines for feature categorization [15]. Even under tough conditions such as changing lighting, background rotation, multiple backgrounds, and different facial regions, the algorithm achieved outstanding accuracy in the range of 85-95%. Performance studies, which included deployment recognition rates and person count charts, demonstrated the LBPH algorithm's superiority over competitors. Bhoyar et al. proposed robust solutions for face detection and identification that were both accurate and applicable in a wide range of settings.

A study focused on the processing speed issues with popular facial recognition algorithms such as R-CNN and Fast R-CNN was proposed by Menon et al. [17]. The YOLO.V3 algorithm was highlighted in the study as a potential means of enhancing processing speed due to its ability to provide speedier outputs, frequently in the millisecond range. The implementation contained a face recognition comparison between YOLO.V3 and R-CNN. The results indicated that although YOLO.V3 processed data quickly, its accuracy was not as high as that of other widely used methods. Hari et al. proposed a facial recognition system based on CNN and compared its performance with established approaches [18]. The work involved gathering datasets, preprocessing data, and training CNN models to identify, align, and recognize faces in real time by comparing their attributes to a database. The results showed that the CNN-based strategy outperformed traditional methods in terms of accuracy and efficiency. This work highlighted the benefits of CNN-based facial recognition systems and promoted continuous improvement through large-scale data training in order to improve prediction. The proposed method by Gowda et al. comprehended the prior efforts in developing student attendance systems by employing face recognition technology [27]. This study undertook a review of existing literature pertaining to student attendance management systems using facial recognition. It provided detailed insights into the methodologies, strategies, and obstacles associated with the application of face recognition techniques. Research focusing on face recognition acknowledged a number of difficulties. In response to these challenges, this study proposed a camera-based student attendance system as face recognition technology.

11 Koodalsamy et al. built a face recognition system using Python's OpenCV module and deep learning techniques [19]. The approach used Haar cascade methods to recognize faces and facial features were retrieved using a combination of CNN methods and the LBPH algorithm. It had a useful attendance tracking tool that marked a face as present based on authentic check-in and check-out timings; otherwise, the face was tagged as unknown. The implementation was efficiently built in Python and thoroughly evaluated on the Raspberry Pi 3 B, providing successful real-time face detection and recognition inside video frames. Notably, the study demonstrated the efficacy of integrating ResNet, SVM, and Haar Cascade approaches to improve the accuracy of face recognition systems. The work done in [19] provided a comprehensive and practical solution for effective facial recognition and attendance management similar to the system proposed by Kumar et al. [26]. It comprised four stages: database creation, face detection, face recognition, and attendance updates. Initially, databases were populated with images of students attending classes. The Haar-Cascade classifier and the Local Binary Pattern Histogram method were employed for face detection and recognition, respectively. This enabled the system to identify and distinguish faces in real-time video feeds from classrooms.

23 Chen et al. developed an iris and facial biometric classifier based on wavelet probabilistic neural networks [20]. The ID wavelet transform and ID energy signal were the features extracted from the iris and face. To create a single feature vector, the study combined two matching scores from an iris verifier based on phase information and a Laplacian face verifier. These features were fused together using a fusion rule based on SVM. By incorporating wavelet techniques and probabilistic neural networks, this method aimed to improve the reliability and precision of face and iris biometric recognition, presenting a comprehensive and unified approach to enhance overall system performance. Zhang et al. tackled the challenge of integrating face and iris biometrics in scenarios with near-infrared lighting, using a single sensor [21]. Frontal face images were captured through a 10-megapixel CCD camera. Local Bit Pattern histogram matching, as outlined in Li et al. [22], was applied for eye detection and face alignment. Face and iris recognition used the eigenface algorithm and Daugman's algorithm, respectively. The integration of scores occurred at the score level, employing sum and product rules post min-max normalization. This strategy offered a comprehensive approach to combine face and iris biometrics in specific lighting conditions, emphasizing the use of distinct algorithms for effective recognition in each biometric modality while ensuring originality.

6 Komlavi et al. conducted a study of facial recognition techniques, with a focus on machine learning and deep learning methodologies [23]. The study assessed and analysed several classification techniques, including SVM, Artificial Neural Networks (ANN), K-Nearest Neighbours (KNN), Random Forests (RF), Logistic Regression (LR), Naive Bayesian Classification (NB), and CNN. The evaluation criteria included the recognition rate, the confusion matrix, and the region under the Receiver Operating Characteristic (ROC) curve. The results showed that CNN regularly outperformed alternative algorithms, especially on the ORL face database. Classical methods such as artificial neural networks performed well on the YALE database. Discussions focused on the effectiveness of deep learning approaches and advocated using picture augmentation to remedy problems. The study concluded that CNN was the most effective classification method for facial recognition, exhibiting better results than the most advanced techniques, with classifier performances reaching 94.82% to 98.95%. The proposed system by Ashwin Rao [25], AttenFace, offered a novel solution to the cumbersome and time-consuming process of marking attendance in educational institutions. Through the use of face recognition technology, AttenFace aimed to streamline attendance tracking by analyzing, tracking, and granting attendance in real-time. By capturing snapshots of classrooms through live camera feeds, the system identified students and recorded their presence based on multiple snapshots taken throughout the class period. This approach allowed for independent and parallel face recognition for each class, ensuring scalability with the number of concurrent classes. Moreover, the separation of the face recognition server from the backend server for attendance calculation enabled seamless integration with existing attendance tracking software such as Moodle. The system's face recognition algorithm ran at intervals of 10 minutes on classroom snapshots, significantly reducing computational load when compared to processing live camera feeds directly. Notably, AttenFace also accommodated students' flexibility to briefly leave class without losing attendance, as attendance was granted based on remaining in class for a specified number of snapshots above a certain threshold.

33 Beldi and Makrem focused on accurately identifying and recognizing human faces for various purposes using machine learning algorithms in [24]. The aim was to develop an accessible Python-based framework for face recognition, promoting innovation in the field. Using TensorFlow and Python, the proposed work refined a CNN model based on the AT&T dataset, achieving outstanding accuracy. The model's performance was further enhanced by careful layer additions. The study recognized the importance of accuracy in facial recognition and underlined the significance of deployment time. Project discussions examined the subtle balance required

between accuracy, operational efficiency, resource allocation, and other aspects involved in implementing effective face recognition.

The Proposed DL-based Secure Real-Time Face Recognition System

In the domain of Android GUI development using the PyQt designer tool, the paper focused on crafting a user-friendly application tailored for facial image capture. This application seamlessly acquired facial images, integrating smooth camera functionality and user prompts to enhance user engagement. The face acquisition module integrated with camera functionality, enabling users to capture facial images directly in the Android app. Techniques ensuring optimal image quality, such as guidance on proper lighting and face positioning, were implemented to elevate the overall user experience. Real-time face detection was achieved through the integration of the Viola-Jones algorithm, by identifying and localizing faces in captured images. The flowchart of the proposed face recognition system was illustrated in Figure 1. The facial encryption module was implemented using the Chaotic Logistic Mapping algorithm to ensure security during the encryption and decryption process. This module transformed pixel values within observed facial images for robust encryption while preserving vital features. Simultaneously, a CNN architecture was constructed specifically for facial recognition. Pre-processed facial images were used to train the CNN model on a labelled dataset stored in a PostgreSQL database. This dual approach addressed the need for secure face encryption and the construction of a successful CNN-based facial recognition model, enhancing the sophistication and functionality.

The backend system was developed using PostgreSQL, which served as the relational database, providing a robust foundation for storing crucial data. Tables were meticulously designed to store user information and their attendance details, ensuring a comprehensive and organized storage structure. Moving to the recognition and verification phase, logic for facial recognition based on the trained CNN model was implemented. Verification processes involved comparing recognized faces with information stored in the database to ensure accuracy and reliability. The user experience was prioritized through the incorporation of user feedback mechanisms during both facial acquisition and recognition processes. The Graphical User Interface (GUI) was enhanced to display recognition results and relevant information in real-time, facilitating a seamless and informative interaction for users. Continuous updates to the database records ensured the system remained accurate and up to date with the latest attendance results.

3.1 Viola Jones Detection

The Viola-Jones Object Detection system, introduced by Paul Viola and Michael Jones in 2001, was the first framework to provide competitive object detection rates. It could detect objects in real time, although it was most commonly employed for applications. This framework's detection rate was surprisingly high, i.e., the true positive rate was high and the false positive rate was low, resulting in a robust algorithm that could handle images rapidly. The primary goal was face detection rather than recognition, which involved distinguishing faces from non-faces as the initial step. The Viola-Jones algorithm consisted of the following distinct modules:

Haar-like Feature: The first being Haar-like Feature Extraction, this module utilized rectangular patterns known as Haar-like features to represent different object aspects, including edges, corners, and lines. Some human faces shared characteristics such as upper cheeks being lighter in hue than the area around the eyes, and the nose bridge receiving lighter than the eyes did. The arrangement of qualities that contributed to distinctive facial features, including the positions and size of the eyes, mouth, and nose bridge. Gradients of pixel intensities were oriented towards value. The image was turned into a window that was 24 by 24. Applied each Haar feature to a window pixel. Every feature was associated with a specific location within the sub-window.

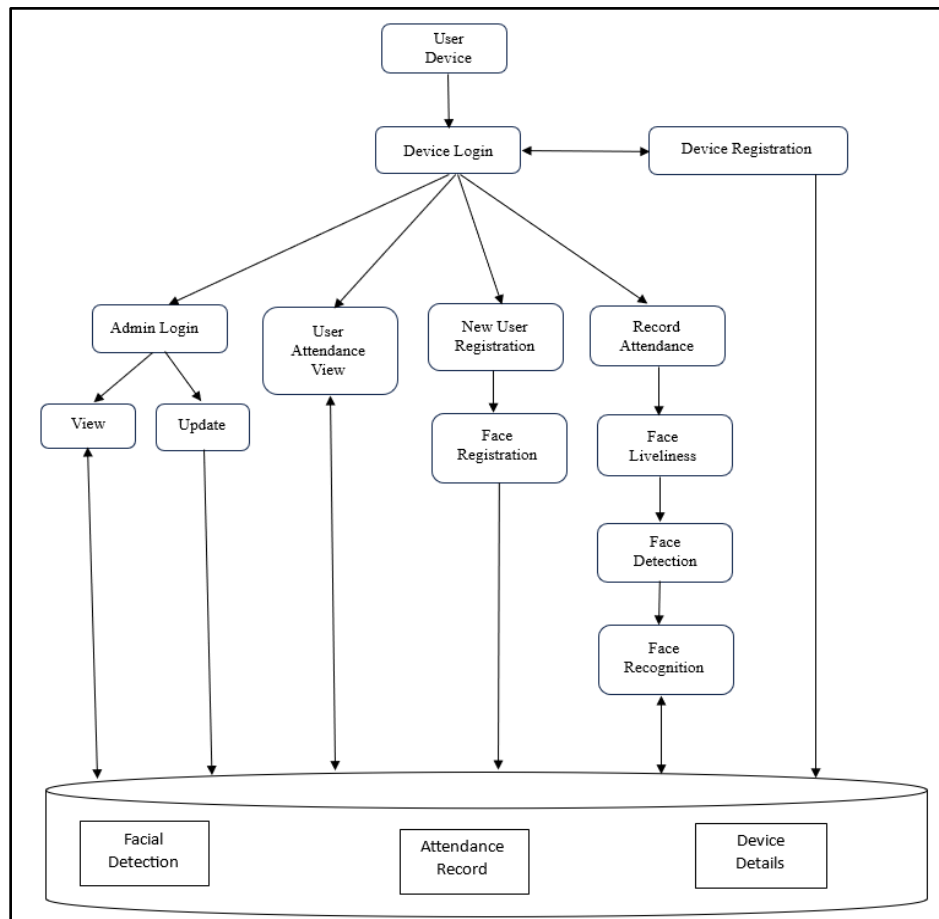


Figure 1. Flowchart of the Proposed Secure Face Recognition System

There were numerous principles for using features rather than pixels directly. Haar's feature-based method had a large performance advantage over pixel-based solutions. The simple features used were similar to Haar basis functions, which were commonly used for pixel recognition. The Haar feature-based technique incorporated three types of features. The value of a two-rectangle feature was derived by subtracting the sum of pixels contained inside two rectangular parts. The zones were identical in both dimension and form and might have been horizontally or vertically contiguous. Figure 2 showed the manipulation of a rectangular region. A three-rectangle feature computed the sum of two outside rectangles reduced by a central rectangle. The four-rectangle feature determined the difference between diagonal pairs of rectangles.

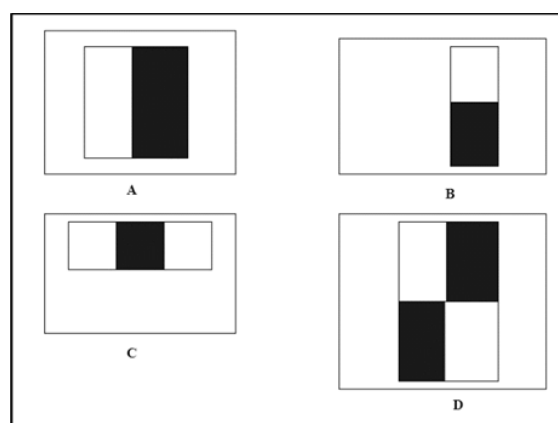


Figure 2Representation of a Haar-Like Features

Integral Image The algorithm aimed to transform a given image into an integral image. The basic resolution of the edge detector could have ranged from 24 x 24, 8 x 8, to 16 x 16. Rectangle features were efficiently computed using an intermediary format called the integral image. At any point (x, y) in the image, the integral image represented the sum of pixels to the upper left of (x, y), inclusive. The computation of rectangle features was expedited by using the integral image as an intermediary representation. The integral image at a given location x, y encapsulated the cumulative sum of pixels positioned above and to the left of the specified coordinates, encompassing an inclusive range, where ii (x, y) was the integral image and point (x', y') was the original image.

$$ii(x, y) = \sum_{\substack{x' \leq x \\ y' \leq y}} I(x', y') \quad \text{-----(1)}$$

$$ii(x, y) = I(x, y) + ii(x, y - 1) + ii(x - 1, y) - ii(x - 1, y - 1) \quad \text{----(2)}$$

Adaboost Learning: AdaBoost is a learning method that combines a series of weak binary classifiers to create a powerful new model which is extensively used for training. The result will have almost 160,000 features, including every single feature. Over 160,000 characteristics, these attributes are used to produce a weighted form. The arrangement of all the qualities used for measurement determines whether a window has a face. These traits are known as weak classifiers. This method extracts important minimal visual features from an immense number of features.

Cascading Classifiers: The last most crucial module within the Viola-Jones algorithm was the Cascade Classifier. This classifier was structured as a cascade, organized into stages. The cascade classifier played a pivotal role in the algorithm's efficiency by rapidly eliminating non-face regions. This cascading structure enhanced the algorithm's speed and efficiency in face detection, contributing to its overall effectiveness. Haar's rectangular characteristics were applied to the face to determine the difference. It scanned the grayscale image to distinguish between black and white pixels. If a face was found, a rectangle was outlined. Many classifiers were pre-trained based on face and eyes, which were already available in OpenCV and XML files. The pseudo-code below illustrated how the algorithm helped in face detection with liveness.

3.2 Convolutional Neural Network

One distinct type of deep learning technique was CNN, which could accept picture as input and identify features or objects that could be consumed by machine learning algorithm to identify and separate one image from another [23]. The architecture of a CNN resembled the organization of neural connections within the human brain. In the architecture of a CNN, the following diverse components played integral roles:

Convolutional Layer: These layers utilized convolutional operations to identify spatial patterns, edges, and features within input data. By sliding convolutional filters over the input, relevant features were captured, contributing to the creation of feature maps. The convolution operation implemented edge zeroing for the input image, with the extent of the edge determined by the size of the convolution kernel. The purpose of edge zeroing was to guarantee logical outcomes, where the elements of the input image and the convolution kernel were sequentially weighted and summed. The convolution kernel was also turned around and flipped up and down, as illustrated in Figure 4, and the kernel was rotated by 180 degrees across its center.

Pooling Layer: These components, also known as subsampling or downsampling layers, reduced the spatial dimensions of convolutional layer-generated feature maps. Popular pooling methods included max pooling, which extracted the maximum value from a group, and average pooling, which calculated the average as shown in Figure 5.

Fully Connected Layers: In the later stages of the network, these layers connected each neuron in one layer to every neuron in the subsequent layer. This facilitated the learning of high-level feature representations, enabling complex classifications.

Activation Layer: These functions introduced nonlinearities into the network, allowing it to capture intricate relationships within the data. Examples of activation functions included Rectified Linear Unit (ReLU), Sigmoid, and Hyperbolic Tangent (Tanh), which were expressed in equations (3), (4), and (5) respectively.

Batch Normalization: This module standardized the inputs of a layer by adjusting and scaling the activations. By improving convergence and stability, batch normalization accelerated the training process, enhancing the efficiency of deep networks.

Algorithm 1 Viola-Jones Face Detection Algorithm

```

Input: Original test image,
Output: Image with face indicated in frame
begin:
for i=1 to num of scales in pyramid of images do
{
  Downsample image to create images
  Compute integral image
  for j=1 to num of shift steps of sub-window do
  {
    for k= -1 to num of filters of stages in cascade classifier do
    {
      for l=1 to num of filters of stage k do Filter detection sub-window
      {
        Accumulate filter outputs
      }
      if accumulation fails per stage threshold
      {
        Reject sub-window as face Break this k for loop
      }
    }
    if sub-window passed all per-stage checks
    {
      Accept this sub-window as a face
    }
  }
}
end
  
```

Figure 3: Pseudo Code of Viola Jones Detection

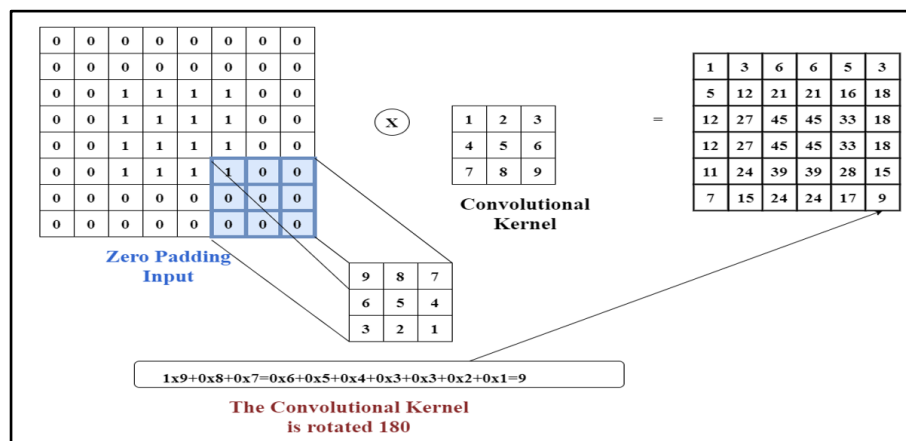


Figure 4 Implementation of Convolutional Layer

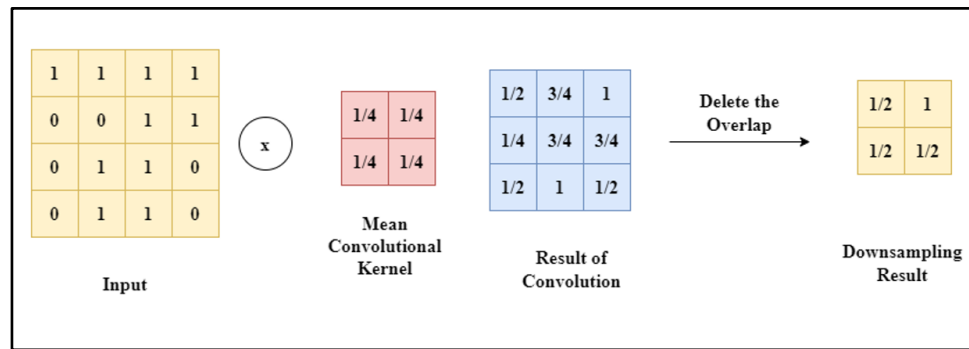


Figure 5 Pooling Operation

$$\text{ReLU} : f(x) = \max(0, x). \quad \text{----- (3)}$$

$$\text{sigmoid} : f(x) = \frac{1}{1+e^{-x}} \quad \text{----- (4)}$$

$$\text{tanh} : f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}, \quad \text{----- (5)}$$

Dropout: As a regularization technique, dropout randomly deactivated a portion of neurons during training. This prevented overfitting by introducing redundancy and diversity, fostering better generalization on unseen data.

CNN Model for Face Recognition: The research work covered the CNN model to increase the categorization accuracy of faces in images. While the model's structure and that of the classical LeNet-5 model were similar, there were differences in the model's input data, network breadth, and entire connection layer. The developed CNN comprised two convolutional layers, namely C1 and C2, and two pooling layers, namely S1 and S2. The normalized facial image was fed into the CNN model using a single feature map in the input layer. The initial convolutional layer, C1, consisted of six feature maps, where each neuron was convolved with a randomly generated 5×5 convolution kernel. Six feature maps were created by the first pooling layer, i.e., S1, which was based on the results of the layer before it.

Data Augmentation: A database of images of faces called LFW (Labeled Faces in the Wild) was created to investigate the issue of unrestricted face recognition. LFW was made available for research purposes to advance face verification rather than to conduct a comprehensive assessment of commercial algorithms before becoming available. The database was an early effort to offer a collection of faces that were categorized and represented a variety of situations that individuals frequently came across in their daily lives. The database demonstrated "normal" variation in postures, lighting, focus, accuracy, facial expressions, age, gender, ethnicity, accessories, cosmetics, occlusions, background, and photography quality. Despite displaying the visuals plainly and consistently for maximum use, there was still a lack of images to properly train the deep neural network for reliable facial recognition. As demonstrated in Figure 7, four data augmentation techniques—horizontal flip, shift, scaling, and rotation—were used to solve this issue by increasing the image amount of the dataset.

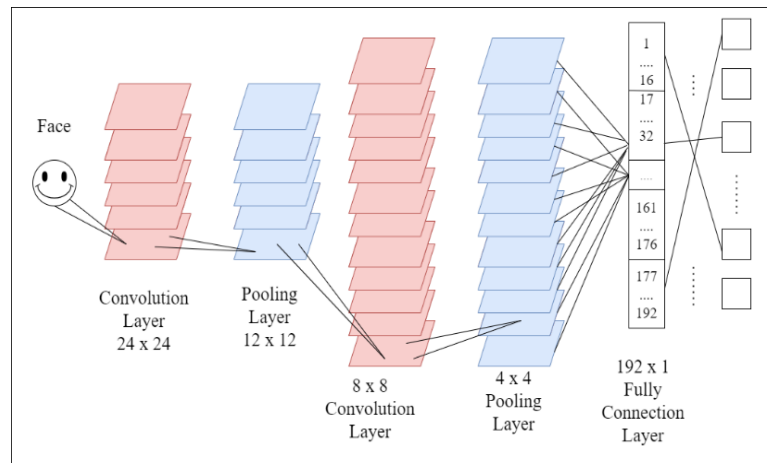


Figure 6 CNN Representation Model for Face Recognition

Expanding a small image dataset into a larger one through image transformation was a practical approach to overcome the challenge of obtaining a large original dataset for facial recognition. This method allowed for the extraction of additional image features, enhancing the training of the classifier and ultimately improving the accuracy of facial recognition results. Following the described operations, the dataset experienced a substantial twenty-fold augmentation as shown in Figure 8.

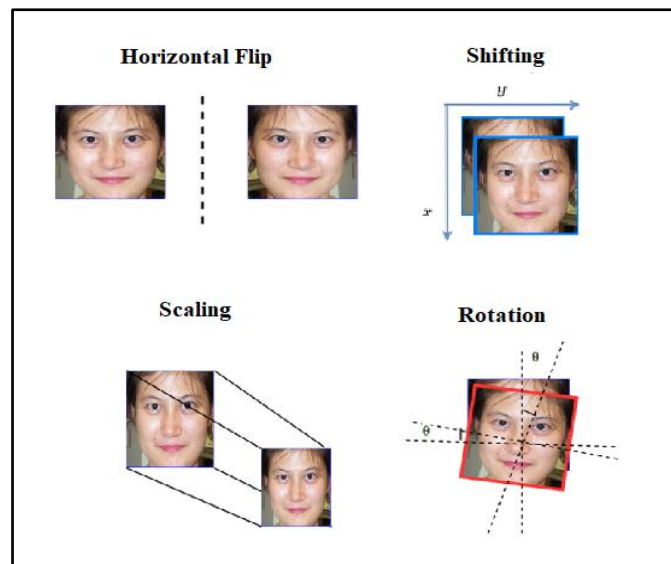


Figure 7 Methods of Augmentation

3.3 Chaotic Logistic Mapping

In chaos theory and dynamical systems, a mathematical function called the logistic equation, sometimes referred to as the logistic map, was frequently employed. Under some parameter values, this straightforward non-linear difference equation displayed complex and chaotic behaviour. Numerous fields, such as population dynamics, encryption, and random number generation, frequently employed the logistic equation.

Pixel Representation: In a color image, each pixel was represented by a combination of three-color channels: red (R), green (G), and blue (B). This approach enabled precise definition of the color of each pixel. For instance, a pixel could be depicted as (R, G, B), where each letter denotes the intensity of Red, Green and Blue colours. Combining these three colored channels enabled us to show a wide range of colors in the image. This

pixel representation was required for manipulating and processing color images in a variety of applications, including image encryption, in which each color channel might have undergone separate encryption operations.

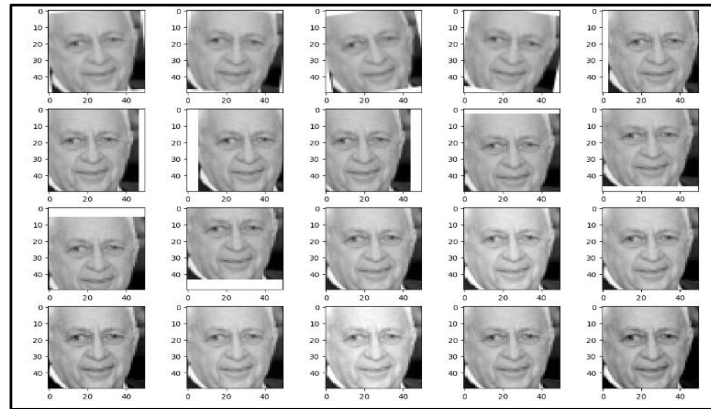


Figure 8 Representation of 20-fold Image Augmentation

Key Generation: The logistic map equation was utilised to produce pseudo-random keys for each colour channel of each pixel during the key generation process. This equation generated a succession of numbers with chaotic and unexpected behaviour, making them useful for encryption purposes. Three keys were generated for each pixel in the image: Kr, Kg, and Kb. These keys represented the red, green, and blue channels, respectively. These keys were used to encrypt the pixels' associated color channels. The encryption procedure offered strong security while retaining the image's color information by creating unique keys for each color channel of each pixel.

XOR Operation and Substitution: In the XOR operation for each color channel of a pixel, the intensity value of the channel was XORed with the corresponding key value, resulting in the encrypted intensity value. Specifically, for the red channel, the intensity value R was XORed with Kr and generated Er. Similarly, the intensity value of the green channel (G) was XORed with Kg and gave Eg, and the blue channel value (B) XORed with Kb yielded Eb. After encrypting each color channel of the pixel, the encrypted values were combined to form the encrypted pixel.

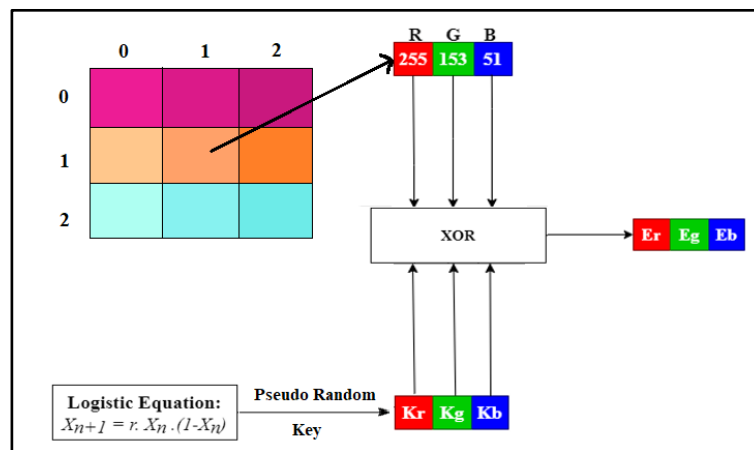


Figure 9 Representation of Chaotic Logistic Mapping Encryption

This method produced an encrypted pixel, with each color channel encrypted independently using its matching key value. The original pixel's color information was intact, but the pixel values were scrambled as a result of the XOR operation on the keys as illustrated in Figure 9. This method was repeated for each pixel in the image, ensuring that each pixel was encrypted separately with a unique set of keys before returning the encrypted image.

Retrieving Key: First, the decryption method recovered the same keys that were used for encryption, (Kr, Kg, Kb). These keys were normally safely stored or transferred with the encrypted image. Here, the keys generated were saved in the file, which helped to retrieve the original image when needed.

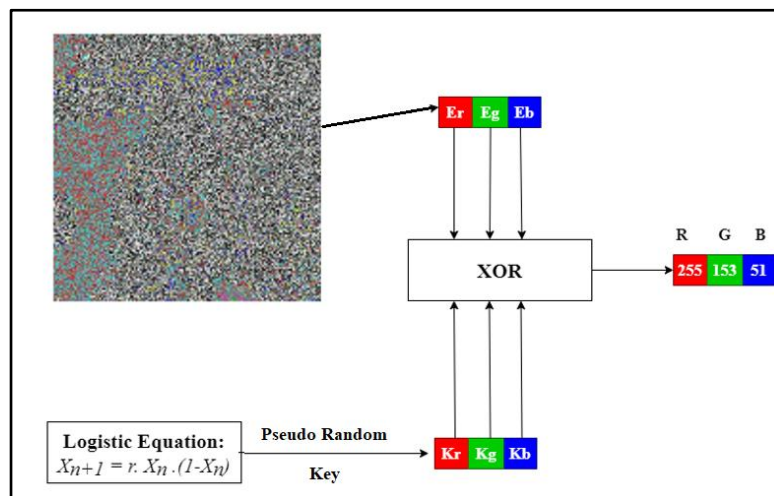


Figure 10 Representation of Chaotic Logistic Mapping Decryption

Reverse XOR Operation: The decryption algorithm reversed the XOR operation on each pixel in the encrypted image, using the associated key values. This procedure generally reversed the XOR technique used during encryption, restoring the intensity values for each color channel in the pixel. In the Red channel, the encrypted value E_r was XORed with K_r and produced the original intensity value, R . Similarly, the encrypted value of the green channel (E_g) was XORed with K_g and yielded G , and the blue channel value (E_b) XORed with K_b yielded B . After decrypting every one of the color channels of the pixel, the results were merged to create the original pixel as shown in Figure 10.

Reconstruction of Original Image: As the XOR operation was carried out on each pixel of the encrypted image with the related key values, the original intensity levels for the red, green, and blue channels were restored. This technique reconstructed the image's original color information, decrypting it. The decryption process returned the original image with its color information restored. This image was identical to the one before encryption, allowing for visual inspection or additional processing as needed.

4 EXPERIMENTAL RESULTS

Models/Metrics	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
LBPH Algorithm	85	86	84	85
SVM +LBPH+Open CV	97	98	97	98
Eigen Face	97	97	97	97
LBPH + Open CV	88	88	87	87
CNN	98	97	98	97

Table 1 Comparison results of various algorithms of Face Recognition

The comparison of various models of face recognition models such as LBPH, Eigen Faces, OpenCV, SVM, CNN and some combined models is shown in Table 1 using accuracy, precision, recall and f1 score. F1 score determines the performance of the model and accuracy determines how well the algorithm predicts the person. The Local Binary Pattern Histogram algorithm has accuracy of 85% with good precision and f1 score. This model has the least accuracy as compared to other models. Combined model of SVM classifier, LBPH and OpenCV gives accuracy of 97% with approximately 98% of precision and f1 score. Using the

Eigen Face algorithm for detection it produced 97% of accuracy, precision, recall and f1 score. The combined algorithm of LBPH and OpenCV has accuracy and f1 score of 88% and 87% respectively. The widely used CNN produces the best accuracy, precision, recall and f1 score around 97-98%. So from the above table comparison, CNN can provide better predictions as compared to other models.

Data split ratio (train data:test data)	Accuracy	Precision	Recall	F1-score
60:40	90	89	89	90
70:30	92	91	92	92
80:20	95	94	94	95
90:10	98	97	98	97

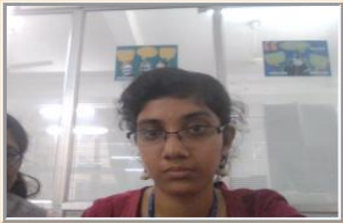
Table 2 Comparison results of different test and train data split using CNN algorithm

Splitting data was crucial for the Face Recognition System as it aided in training, testing, and validating machine learning models. This practice ensured unbiased evaluation, prevented overfitting, supported tuning model parameters, allowed model comparison, and helped in identifying the better approach to provide prediction. Table 2 depicted the comparison results based on different test and train data splits over the CNN algorithm. The data set on splitting in a 60:40 ratio gave an accuracy of 90% and with good precision and F1 score. The data split ratio of 70:30 gave better accuracy, precision, and F1 score. So, improving the dataset ratio of training data improved the accuracy and F1 score. From the comparison, it was evident that the data split of a 90:10 ratio produced better accuracy and predictions.

The application used image analysis to detect individuals in real time and logged the resulting updates to the database. Figure 11 is the user interface for attendance recording, where users could select either the check-in or check-out button. After clicking, video capturing began, first determining whether the observed face was genuine or faked. Following that, facial recognition began, and attendance records were updated accordingly.

Take Attendance

Check InCheck Out



Username:

Varshini

Date:

2024-03-14

Time:

11:12:15

Ok

Figure 11 Attendance Recording after clicking the check in or check out button.

If the system recognized a fake face, as shown in Figure 12, the procedure is halted to prevent erroneous attendance entries from being generated, assuring data accuracy and integrity. A separate user login was maintained, which displays the user details and other details such as attendance percentage, total number of working days, and the check-in and check-out time details in the table shown in Figure 13.

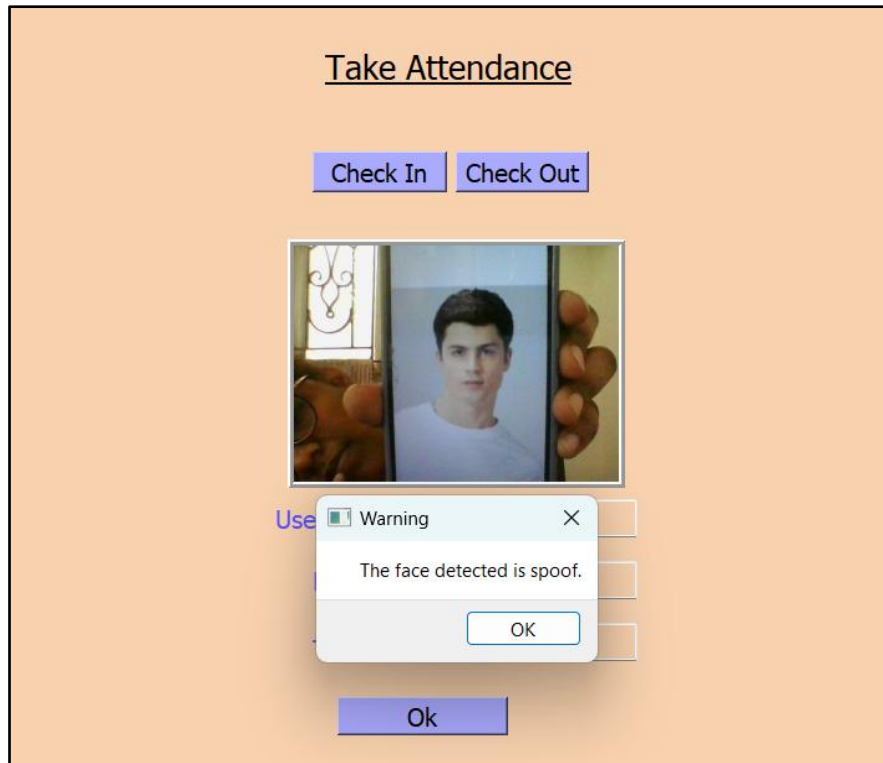


Figure 12 Detection of spoof image

Details

Name:

User ID:

Department:

Email ID:

Attendance Details

No. of working days:

Attendance Percent:

Days Present

	Date	Check In Time
5	2024-04-03	08:19:00
6	2024-04-04	08:19:00
7	2024-04-05	08:19:00

Figure 13 User Attendance Details View

4 CONCLUSIONS AND FUTURE WORK

The research work successfully demonstrated the efficient deep learning-based attendance system with a custom-trained model for anti-spoofing and face recognition, outperforming other algorithms such as LPBH, SVM, and Eigen Face in terms of precision and recall. The integration of front-end development using PYQT enhanced user interaction and visualization of results, making the system accessible and user-friendly. The details were recorded with the help of PostgreSQL. This project ensured security and efficiency for performing attendance.

Looking back, future work focused on several areas to further advance the proposed idea's capabilities and applicability. Firstly, there was scope for performance optimization of the CNN algorithm for multiclass and custom-trained models to improve real-time processing speed and resource efficiency, enabling faster inference without compromising accuracy. Secondly, the integration of the cloud where the model could train new users and store face registry images was considered. Additionally, continued research into algorithm comparison, deployment, and scalability aspects was crucial for ensuring the system's viability and scalability in real-world deployment scenarios. By addressing these areas, it continued to evolve and made significant contributions that improved accuracy and response time for recognizing faces to record attendance.

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